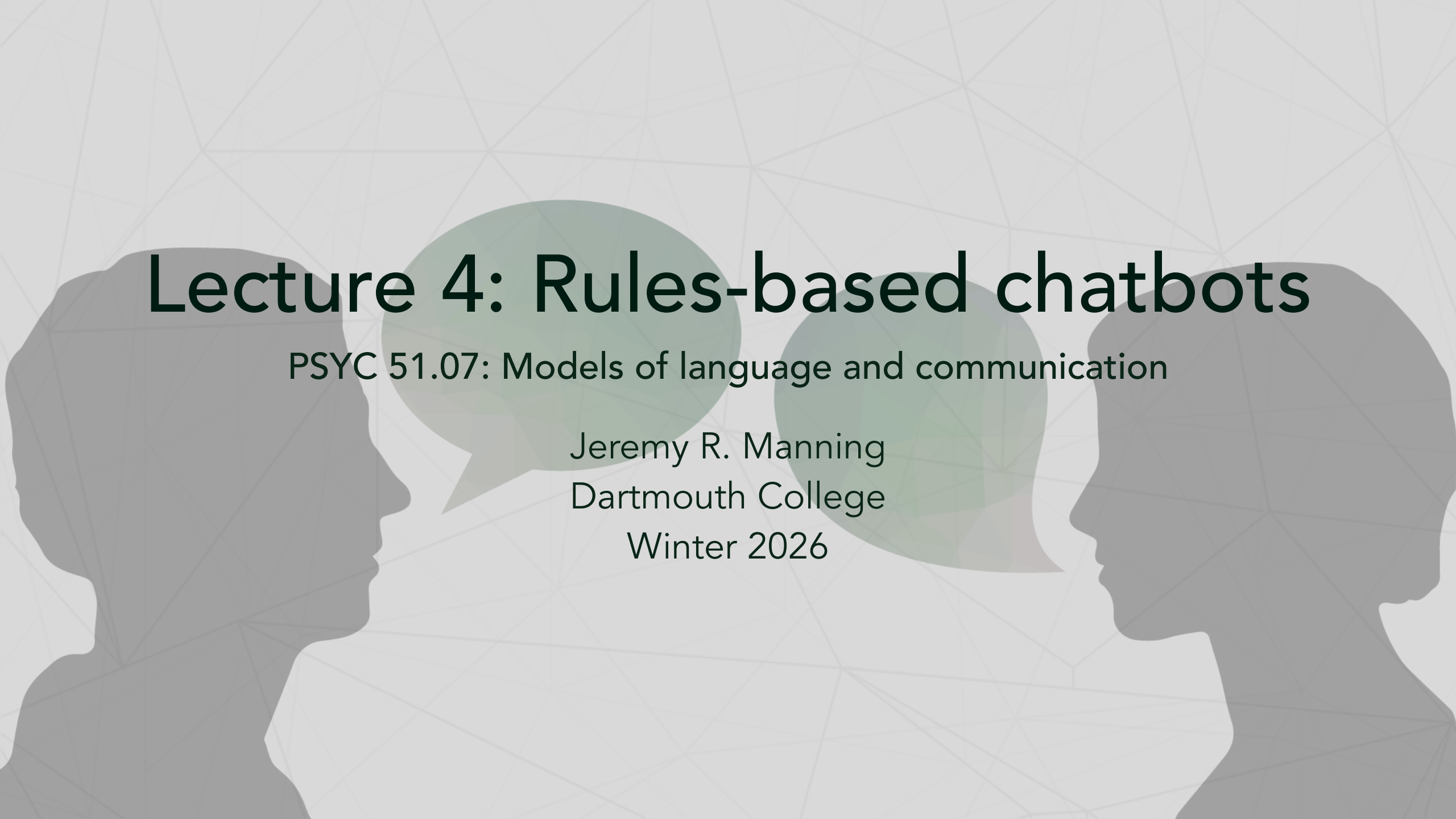




Lecture 4: Rules-based chatbots

PSYC 51.07: Models of language and communication

Jeremy R. Manning
Dartmouth College
Winter 2026

Assignment 1: Q&A

Questions welcome

Let's take some time to address any questions about Assignment 1:

- Implementation challenges
- Pattern matching strategies
- Configuration file format
- Testing approaches

Remember

There are no bad questions. If you're confused about something, others probably are too.

Common issues and tips

Technical tips

- Use raw strings for regex: `r"pattern"`
- Test patterns on regex101.com
- Handle edge cases (empty input, special chars)
- Print intermediate results for debugging

Common pitfalls

- Forgetting pre-substitutions before matching
- Case sensitivity issues
- Greedy vs. non-greedy matching
- Substitution order matters

Beyond ELIZA

A foundation for more

ELIZA (1966) was just the beginning. Weizenbaum's ideas inspired other researchers to explore what rules-based systems could do.

Today we'll explore:

- **PARRY** (1972): A different kind of simulation
- **A.L.I.C.E.** (1995): Pattern matching at scale
- **Formal grammars**: The theoretical foundations
- The limits of rules-based approaches

PARRY (1972)

Further reading

Colby, Weber, & Hilf (1971, *Artificial Intelligence*): Artificial Paranoia

- Created by psychiatrist **Kenneth Colby** at Stanford
- Simulated a patient with **paranoid schizophrenia**
- Different goal than ELIZA: model a specific mental illness
- Had internal state: beliefs, emotional level, goals

Key insight

ELIZA *reflects*; PARRY *models*. ELIZA avoids commitment; PARRY has a coherent (if paranoid) worldview.

PARRY vs ELIZA

ELIZA (1966)

- Non-directive therapy
- Reflects user input back
- No internal state or beliefs
- Avoids making claims
- Goal: keep user talking

PARRY (1972)

- Simulates a patient
- Has beliefs about the world
- Tracks emotional "anger" level
- Makes paranoid claims
- Goal: behave like a specific illness

Important caveat

Both are still rule-based systems with no real understanding. The "beliefs" in PARRY are just variables that affect pattern selection.

The PARRY algorithm

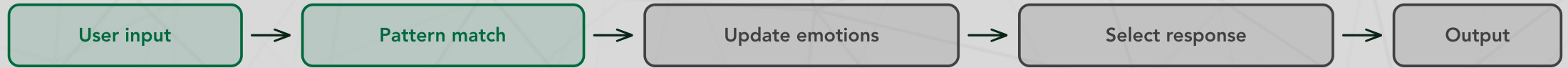


PARRY's processing pipeline with emotional state

Key difference from ELIZA

PARRY maintains **persistent emotional state** across turns. This state influences which responses are selected, creating coherent paranoid behavior over time.

Pattern matching: detect triggers in user input



How does it work?

PARRY scans input for **trigger keywords** organized by topic. Each topic has associated emotional effects and response pools.

Example trigger categories

Category	Keywords	Emotional effect
Mafia/mob	"mafia", "mob", "gangster"	Fear +4, Mistrust +5
Police	"police", "cop", "arrest"	Mistrust +4, Anger +3
Trust	"trust", "believe", "honest"	Mistrust +3
Racetrack	"horses", "racing", "track"	Anger -1 (calming)

Emotional update: modify internal state

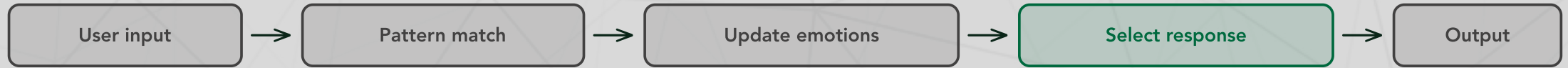


How does it work?

When a trigger pattern matches, PARRY adjusts its emotional variables. These values persist across the conversation.

```
1  class Parry:
2      def __init__(self):
3          self.anger = 5          # 0-20 scale
4          self.fear = 8           # 0-20 scale
5          self.mistrust = 10      # 0-15 scale
6
7      def process_trigger(self, topic):
8          if topic == "mafia":
9              self.fear += 4
10             self.mistrust += 5
```

Response selection: choose based on emotional state



How does it work?

PARRY selects responses from different pools based on current emotional thresholds. Higher emotions trigger more paranoid responses.

Response pools by emotional state

Emotional level	Response style	Example
Low (calm)	Cooperative	"I used to gamble on horses."
Medium	Guarded	"I don't want to talk about that."
High (paranoid)	Hostile	"Are you one of THEM?"

Worked example: PARRY conversation

Early in conversation (low emotions):

1	anger=5, fear=8, mistrust=10
2	
3	Doctor: Tell me about yourself.
4	PARRY: I went to the races at Bay Meadows.
5	[Neutral topic, cooperative]
6	
7	Doctor: Did you win?
8	PARRY: I bet a little on the horses.
9	[Still calm, sharing]

After triggering topics (high emotions):

1	anger=15, fear=16, mistrust=14
2	
3	Doctor: What about the mafia?
4	PARRY: I know what you're up to!
5	[Defensive, suspicious]
6	
7	Doctor: I'm just trying to help.
8	PARRY: You're trying to get me to
9	say something I don't mean.
10	[Paranoid interpretation]

Key insight

PARRY's responses are shaped by accumulated emotional state, creating a coherent "personality" over time.

The Turing Test, revisited

A famous experiment

In 1972, PARRY was tested via teletype against real patients and psychiatrists. Judges could not reliably distinguish PARRY from actual patients with paranoid schizophrenia.

- This was one of the first informal "Turing tests"
- Success? Or a comment on how we judge understanding?
- Psychiatrists were looking for *symptoms*, not *understanding*

Think about it

Does passing a specialized test mean the system understands anything? What does it mean that experts could be fooled?

A.L.I.C.E. (1995)

The AIML era

A.L.I.C.E. (Artificial Linguistic Internet Computer Entity) was created by Richard Wallace.

Key innovations:

- **AIML** (Artificial Intelligence Markup Language)
- Over **40,000 patterns** (vs ELIZA's ~200)
- Won the Loebner Prize three times (2000, 2001, 2004)
- Open source, widely studied and extended

Warning

More patterns $\neq$ more understanding. A.L.I.C.E. is still fundamentally pattern matching.

The ALICE algorithm



ALICE's AIML processing pipeline

Key innovation

AIML supports **recursive processing** via `<srai>` (Symbolic Reduction AI), allowing patterns to trigger other patterns. This enables handling many input variations with fewer rules.

Normalization: prepare input for matching



How does it work?

ALICE normalizes input to uppercase and removes punctuation before pattern matching. This reduces the number of patterns needed.

Example normalization

Input	Normalized
"Don't you think so?"	"DO NOT YOU THINK SO"
"What's your name?"	"WHAT IS YOUR NAME"
"I can't believe it!"	"I CAN NOT BELIEVE IT"

Pattern search: find matching AIML categories



How does it work?

ALICE searches through 40,000+ patterns to find the best match. Patterns use wildcards (`*` and `_`) to capture variable text.

```
1 <category>
2   <pattern>MY NAME IS *</pattern>
3   <template>Nice to meet you, <star/>.</template>
4 </category>
5
6 <category>
7   <pattern>I AM FEELING *</pattern>
8   <template>Why are you feeling <star/>?</template>
```

Sound familiar?

This is the same decomposition/reassembly pattern as ELIZA, just in XML format with more extensive coverage.

SRAI: recursive pattern matching



How does it work?

The `<srai>` tag redirects processing to another pattern. This allows many input variations to map to a single response.

```
1  <!-- These all redirect to the same base pattern -->
2  <category>
3    <pattern>HI THERE</pattern>
4    <template><srai>HELLO</srai></template>
5  </category>
6
7  <category>
8    <pattern>HOWDY</pattern>
9    <template><srai>HELLO</srai></template>
10 </category>
11
```

Worked example: AIML recursive processing

Input: "Do you know what my name is"

Step 1: Direct pattern search

```
1 <pattern>DO YOU KNOW WHAT MY NAME  
  IS</pattern>  
2 → No exact match found
```

Step 2: Try wildcard patterns

```
1 <pattern>DO YOU KNOW *</pattern>  
2 <template>  
3   <srai>WHAT IS <star/></srai>  
4 </template>  
5 → MATCH! Redirect to "WHAT IS my  
  name"
```

Step 3: Recursive processing

```
1 <pattern>WHAT IS MY NAME</pattern>  
2 <template>  
3   <condition name="name">  
4     <li value="unknown">  
5       I don't know. What is your  
6       name?  
7     </li>  
8     <li>Your name is <get  
9       name="name" />.</li>  
    </condition>  
  </template>
```

Result: "I don't know. What is your name?"

The power of SRAI

SRAI lets you handle many input variations without writing separate rules for each.

Live demo

Interactive exploration

Let's interact with these historical systems:

[Chatbot Evolution Demo](#)

While exploring, consider:

- How do PARRY and A.L.I.C.E. differ from ELIZA?
- What do they do surprisingly well?
- Where do they break down?
- Can you find the edges of their rule sets?

Formal grammars: the theory behind pattern matching

Further reading

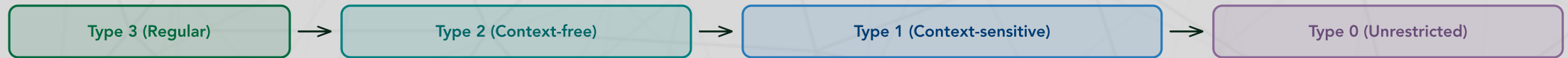
Chomsky (1956): "Three Models for the Description of Language," *IRE Transactions on Information Theory*

In 1956, Noam Chomsky introduced a **hierarchy of formal grammars** that classify languages by the complexity of rules needed to generate them.

Why does this matter?

The Chomsky hierarchy tells us what kinds of patterns different computational systems can recognize—and what they *cannot*.

The Chomsky hierarchy



Each level can express everything below it, plus more

Type	Grammar	Recognizer	Example
Type 3	Regular	Finite automaton	<code>a*b+</code> (any a's followed by b's)
Type 2	Context-free	Pushdown automaton	Balanced parentheses: <code>(())</code>
Type 1	Context-sensitive	Linear-bounded automaton	$a^n b^n c^n$
Type 0	Unrestricted	Turing machine	Any computable language

Key insight

Regular expressions (which ELIZA, PARRY, and ALICE use) are **Type 3**—the simplest class!

Regular expressions = Type 3 grammars

Mathematical equivalence

Regular expressions and Type 3 (regular) grammars are **provably equivalent**—they recognize exactly the same class of languages.

Regular expression

1 | (ab)*c+

Matches: "c", "cc", "abc",
"ababcc"

Equivalent Type 3 grammar

1 | S → A | C

2 | A → abA |
abC

3 | C → c | cC

Chomsky's claims about human language

Historical context

Chomsky argued that human languages require **at least** context-free (Type 2) grammars, and likely context-sensitive (Type 1) or beyond.

What Chomsky showed:

- Human syntax has recursive, nested structure
- Sentences like "The cat the dog chased ran" require tracking dependencies
- Regular grammars (Type 3) are **insufficient** for natural language

The implication:

- Pattern matching with regex can only approximate language
- No matter how many patterns (200 or 40,000), fundamental structures will be missed
- We need more powerful computational models

Preview

This theoretical limitation foreshadows why rule-based chatbots ultimately fail—and why the field moved toward statistical and neural approaches.

Discussion: The limits of rules

Discussion questions

1. More rules $\neq$ understanding. Why not?
2. What *can't* rules-based systems do?
3. Where do they break down?
4. Is there a fundamental limit, or just a practical one?

Consider these failure modes:

- Novel situations not covered by rules
- Context that spans multiple turns
- Reasoning about cause and effect
- Understanding sarcasm, irony, metaphor

Worked example: Where rules fail

Problem 1: Novel combinations

```
1 | User: My robot vacuum ate my  
  | homework  
2 | ALICE: I have no response for that.  
3 |  
4 | # No pattern for this specific  
  | situation  
5 | # Even though it understands  
  | "vacuum"  
6 | # and "homework" separately
```

Problem 2: Multi-turn reasoning

```
1 | User: I bought a red car yesterday  
2 | ALICE: Nice! I like cars.  
3 |  
4 | User: What color is it?  
5 | ALICE: What is what color?  
6 |  
7 | # ALICE forgot the car was red!
```

Problem 3: Implicit meaning

```
1 | User: Can you pass the salt?  
2 | ALICE: Yes, I am capable of passing  
  | salt.  
3 |  
4 | # Literal interpretation misses  
5 | # the REQUEST implied by the  
  | question
```

Problem 4: Sarcasm

```
1 | User: Oh great, another Monday.  
2 | ALICE: What makes Mondays so great  
  | for you?  
3 |  
4 | # Misses the negative sentiment  
  | entirely
```

The failure of rules: a paradigm shift

The harsh reality

Even with **40,000+ hand-crafted patterns**, ALICE cannot:

- Handle novel combinations of known concepts
- Maintain context across a conversation
- Understand implicit meaning or subtext
- Generalize beyond its training examples



More rules don't solve the fundamental problem

The insight that changed everything

What if, instead of *writing* rules by hand, we could *learn* patterns automatically from massive amounts of text data?

From rules to data

Rules-based approach

- Humans write patterns
- Limited by human creativity
- Brittle to variations
- No generalization
- Example: ELIZA, PARRY, ALICE

Data-driven approach

- Learn patterns from data
- Scale with more data
- Handle variations naturally
- Generalize to new inputs
- Example: Neural networks, LLMs

What rules-based systems teach us

1. **Behavior $\neq$ understanding**

- A system can *seem* intelligent without understanding anything
- This is the lesson of the Chinese Room

2. **Scale is not a solution**

- 40,000 patterns is not qualitatively different from 200
- More rules just mean more edge cases

3. **Pattern matching can fool us**

- We project understanding onto systems that lack it
- The ELIZA effect applies to all of these systems

Weizenbaum's legacy

Further reading

Weizenbaum, J. (1976). *Computer Power and Human Reason: From Judgment to Calculation*. W.H. Freeman.

After creating ELIZA, Weizenbaum became a critic of AI:

- Warned against replacing human judgment with computation
- Argued some tasks *should not* be automated
- Foresaw many issues we face today with AI

"What I had not realized is that extremely short exposures to a relatively simple computer program could induce powerful delusional thinking in quite normal people."

Weizenbaum's warnings

His concerns (1976)

- Mistaking simulation for understanding
- Replacing human relationships
- Trusting AI with decisions it cannot make
- Questions about human identity

Still relevant today

- Chatbots as therapists
- AI companions and relationships
- Automated decision systems
- Debates about AI consciousness

50 years later

Weizenbaum's warnings feel remarkably prescient. Are we heeding them?

Up next...

Week 2: Computational linguistics

We're leaving hand-crafted rules behind. Next week we explore how to **learn from data**:

- **Lecture 5:** Data cleaning and preprocessing
- **Lecture 6:** Tokenization—breaking text into meaningful units
- **Lecture 7:** Text classification workshop
- **Lecture 8:** POS tagging and sentiment analysis

Hand-crafted rules



Learning from data

The paradigm shift that enables modern NLP

The key idea

Key takeaways

1. **PARRY adds emotional state:** Pattern matching + persistent variables = coherent personality
2. **ALICE scales patterns:** 40,000 rules with AIML and recursive SRAI processing
3. **Chomsky hierarchy:** Regular expressions (Type 3) are the *weakest* class of grammars
4. **Fundamental limits:** Rules capture syntax, not semantics—no amount of patterns can bridge this gap
5. **The paradigm shift:** From hand-written rules to learning patterns from data
6. **Weizenbaum's warnings:** The tendency to anthropomorphize remains relevant today

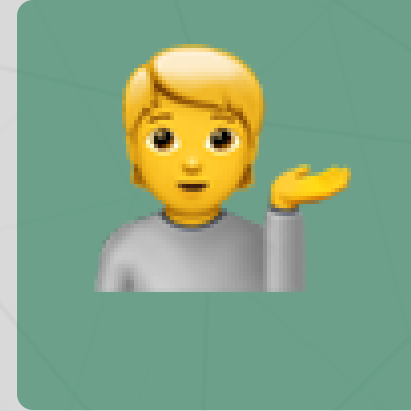
Questions? Want to chat more?



Email me



Join our Discord



Come to office hours

Tip

Start working on Assignment 1 now if you haven't already. Reach out early if you get stuck.